Organic Cumulative Examination
April 9, 1994

1. The following award notices in recent issues of C&EN. Name these famous organic chemists. (10 points each)

a. **Roger Adams Award in Organic Chemistry**

sponsored by Organic Reactions Inc. and Organic Syntheses Inc.

professor at Harvard University since 1959, has been described by a colleague as "the world's undisputed leader in synthetic organic chemistry. He has literally changed the way people think about the problem of creating organic architecture. No one individual, now living, has had such a profound impact on this field."

Winner of the 1990 Nobel Prize in Chemistry, the Japan Prize in Medicinal Sciences in 1989, and the Wolf Prize in Chemistry for 1986, has been contributing to organic and medicinal chemistry since receiving a Ph.D. degree from Massachusetts Institute of Technology in 1950. He has more than 700 publications in scientific journals and has held over 100 special lectureships. Corey is a member of ACS and the National Academy of Sciences; he has received numerous honorary degrees and has honorary memberships in the Royal Society of Chemistry, the Chemical Society of Finland, the Chemical Society of Japan, and the Pharmaceutical Society of Japan.

His research interests include synthesis of complex bioactive molecules; the logic of chemical synthesis; new methods of synthesis; molecular catalysts and robots; theoretical organic chemistry and reaction mechanisms; organometallic chemistry; bioorganic and enzyme chemistry; eicosanoids and their relevance to medicine; and application of computers to organic chemical problems, especially to retrosynthetic analysis.

After receiving a Ph.D. degree, was for several years a member of the chemistry faculty at the University of Illinois, Urbana-Champaign—a job that was offered to him by Roger Adams.

In 1968, published the first of his 50 pioneering contributions to the prostaglandin eicosanoid field. He was the first to synthesize the major representatives of this class of fatty-acid-derived hormones, including prostaglandins and leukotrienes. He has also studied the biosynthesis of the eicosanoids with regard to intermediates and effective inhibition. His work established fundamentally different pathways for prostaglandin biosynthesis in mammalian and marine systems. The design principles and ideas that he developed are now the basis for the pharmaceutical research and production of new therapeutic agents in the eicosanoid field.

also has had a major impact as an educator. During the past 35 years, he collaborated with more than 375 individuals; 82 were students who received Ph.D. degrees under his direction and 295 were either postdoctoral fellows or visiting faculty who spent from one to three years in his laboratory. Of these, 160 hold academic positions in major universities throughout the world. One of his postdoctoral colleagues subsequently was awarded the Nobel Prize in Medicine. According to a colleague, students in research group are educated by their involvement in a wide range of challenging research projects at the frontiers of fundamental chemical science.

b. **Arthur C. Cope Award**

contributions to chemistry during his more than 50 years in the profession extend far beyond the new knowledge generated by his research. His accomplishments have profoundly influenced how chemists think about, approach, and teach chemistry.

work helped shape physical organic chemistry. He was one of the first to use the tools of physical chemistry—quantum mechanics, molecular spectroscopy, kinetics, and other techniques—to solve problems in organic chemistry. "If the idea that organic chemistry and physical chemistry share a common border now seems self-evident, it is in major part because demonstrated the utility of research techniques drawn from the latter in solving problems of the former," writes one of his colleagues.

For example, played a role in developing hydrogen-1, carbon-13, nitrogen-15, and fluorine-19 nuclear magnetic resonance (NMR) spectroscopies as organic chemistry techniques. He used NMR to probe the structure and reactivity of molecules, including nitrogen-inversion, rotation about carbon-nitrogen bonds, rotational isomerism about

single bonds, conformational equilibration and conformational analysis, and enzyme structures.

b. (cont'd)

pioneered the use of isotopic labeling to trace molecular rearrangements. He also introduced organic chemists to the use of molecular-orbital theory for exploring the structure and reactions of organic molecules. And he defined methods of determining structures of reactive intermediates, such as nonclassical cations—a term he coined—and benzyne.

received both B.A. (1941) and Ph.D. (1944) degrees in chemistry from the University of California, Los Angeles. After a postdoctoral stint at Harvard University, he joined the faculty of Massachusetts Institute of Technology. He moved to California Institute of Technology's division of chemistry and chemical engineering in 1953, where he has remained ever since. He has served as Caltech's vice president, provost, and dean of the faculty.

As emeritus Institute Professor of Chemistry since 198[illegible], has continued his NMR studies with undergraduate research students, five of whom worked in his lab this past summer. "Undergraduate research in chemistry changed the course of my life," he says. "Now I'm trying to do what I can to help someone else." He also teaches a course and is writing a book on intermediate NMR concepts for advanced undergraduates and beginning graduate students.

has received myriad awards, including the 1987 Priestley Medal, the American Chemical Society Award in Pure Chemistry, the Roger Adams Award, the Linus Pauling Award, and the Robert A. Welch Award. In 1990 he was awarded the National Medal of Science. He is a member of ACS, the American Philosophical Society, and the National Academy of Sciences.

C.

Alfred Burger Award in Medicinal Chemistry

sponsored by SmithKline Beecham

research professor of chemistry at the University of Pennsylvania, has had a profound impact on chemical research in the biomedical sciences and on both academic and industrial chemistry. He is particularly distinguished for his research in two diverse areas of natural products chemistry—steroids and peptides. All of his research in the biomedical sciences has mechanistic organic chemistry as its basis.

C. (continued)

One of his most notable achievements was the codirection, with Robert G. Denkewalter, of the first successful solution synthesis of an enzyme (ribonuclease), which was announced simultaneously with a solid-phase synthesis by Merrifield in January 1969. These chemical syntheses of active enzymes were achieved at a time when it was still widely believed that the total chemical synthesis of an enzyme without use of a template was not feasible. These two syntheses have been regarded as major milestones in both synthetic organic chemistry and in biochemistry.

In the early 1950s, discovery of the "C-nor D-homo rearrangement" of a steroid enabled him to suggest a biosynthetic pathway that related the veratrum alkaloids, which had just been chemically characterized, to the much better known steroids. At least as important was that in this publication in 1952, explicitly described the rearrangement to be under stereoelectronic control. Another important discovery was the first recognition, jointly with N. L. Wendler, that the direction of enolization of steroid ketones is not solely determined by the degree of substitution. Equally unorthodox was his demonstration with M. Tishler that the 3-keto-4-pregnene structure of steroids is not required for anti-inflammatory activity, in that the [3,2-c] steroid pyrazoles are the most potent glucocorticoids known.

In the 1960s, introduced a novel rational bioorganic concept into medicinal chemistry. He showed that the therapeutic index of a biologically active compound can be improved by drug latentiation; this involved the enzymatic conversion of an inactive prodrug into the active drug preferentially at the desired site in vivo.

In the 1970s, associates, under the direction of D. F. Veber, established the bioactive conformation of somatostatin by a combination of molecular modeling, sophisticated intuition, and creative synthesis. Using this information as a springboard, a cyclic hexapeptide was designed that was completely stable to proteases and therefore possessed a longer biological half-life and that was also more potent than the tetradecapeptide somatostatin.

received an A.B. degree in 1943 from Oberlin College and M.A. and Ph.D. degrees from the University of Wisconsin, Madison, in 1948 and 1950, respectively, under the direction of W. S. Johnson. He joined Merck Research Laboratories in 1950 as a process chemist and advanced steadily until his promotion to senior vice president of basic research in 1978. In 1984, he became senior vice president for chemistry. After retiring from Merck in 1987, at the mandatory age of 65, was appointed research professor of chemistry at the University of Pennsylvania, where he initiated collaborative research with A. B. Smith III and K. C. Nicolaou in 1988 in the areas of peptidomimetics with novel scaffolding, in peptide synthesis, and—in collaboration with Smith and Stephen J. Benkovic at Pennsylvania State University—in antibody research. He holds a concurrent appointment as University Professor of Biomedical Research at the Medical University of South Carolina, Charleston.

2. The following organic molecules all have an important usage; industrial, laboratory, biochemical or pharmaceutical. Give the common name (for 3 points) and the usage (2 points) of any 12 of them. Only your first 12 answers will be graded so avoid wild guessing.

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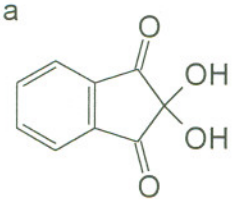

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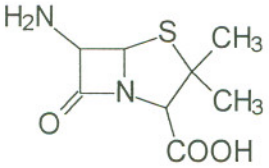

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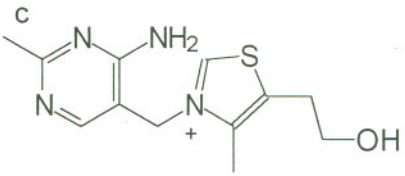

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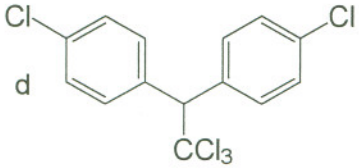

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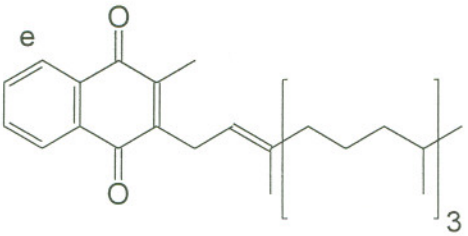

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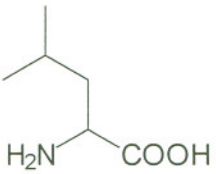

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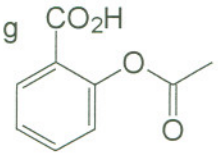

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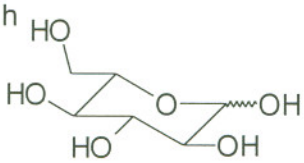

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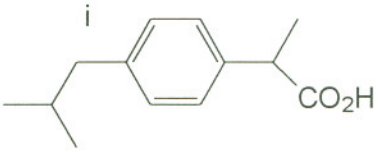

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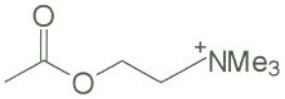

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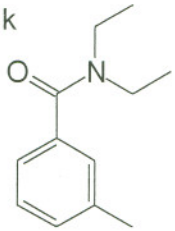

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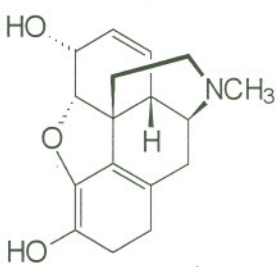

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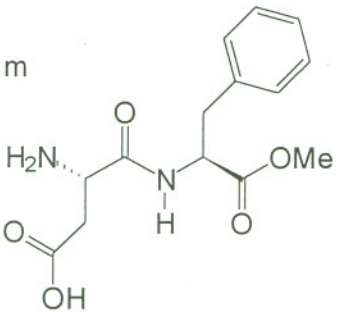

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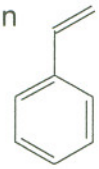

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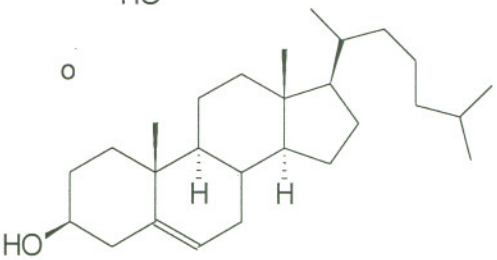

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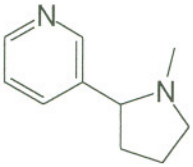

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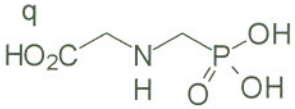

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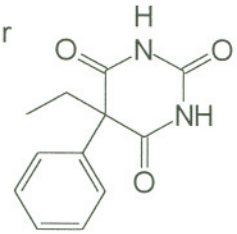

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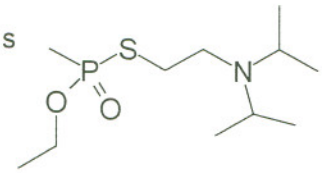

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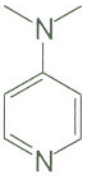

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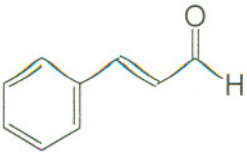

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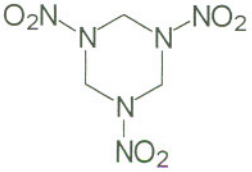

d.

to receive 1994 Nichols Medal

The Nichols Medal Jury of the ACS New York Section has selected Bren Professor of Chemistry at California Institute of Technology, to receive the 1994 William H. Nichols Medal. is being cited for "his outstanding contributions in the field of bioorganic chemistry through the design and synthesis of sequence-specific DNA cleaving molecules."

The award, which consists of a gold medal and a bronze replica, was established in 1902. It honors an investigator who has published a significant and original contribution in the field of chemistry within the past five years.

has greatly influenced the course of organic chemistry through his studies directed toward understanding the chemical principles for the sequence-specific recognition of double helical DNA. He has brought synthesis and design to the field of biopolymers and the methodologies of nucleic acids to the field of molecular recognition. Recent work from his laboratories has demonstrated that quantitative reactions on single atoms at a single site within 15 million base pairs of DNA are possible using strictly chemical methods.

received a B.S. degree in chemistry from Boston College in 1967 and a Ph.D. degree in chemistry from Yale University in 1972. He has received numerous ACS national and local section awards, among them the Nobel Laureate Signature Award for Graduate Education in Chemistry (1985), an Arthur C. Cope Scholar Award (1986), the Harrison Howe Award (1988), the Arthur C. Cope Award (1993), and the Willard Gibbs Medal (1993). is a member of ACS and the National Academy of Sciences and is a fellow of the American Academy of Arts & Sciences.